

Diamond chips, from wafer to qubit

ADAMAS: an open, tested framework for diamond electronics and room-temperature quantum processors

Aleksander Norman

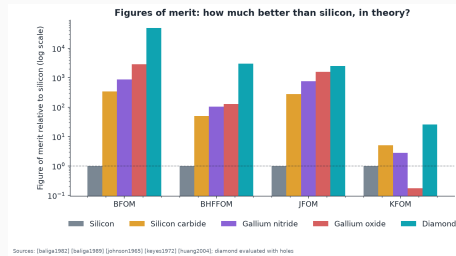
Every number on these slides is generated by the package `(make talk)`

California State University, Dominguez Hills · github.com/Normansrule/adamas-diamond-framework

Why diamond at all?

- Power-switch figure of merit: $48,976\times$ silicon
- Heat removal: $14.7\times$ silicon
- A defect, the nitrogen-vacancy (NV) center, is a **qubit at room temperature**

But: boron dopants are 0.37 eV deep (under 1% active at 300 K), wafers only reached 3 inches recently, and logic is p-channel only.



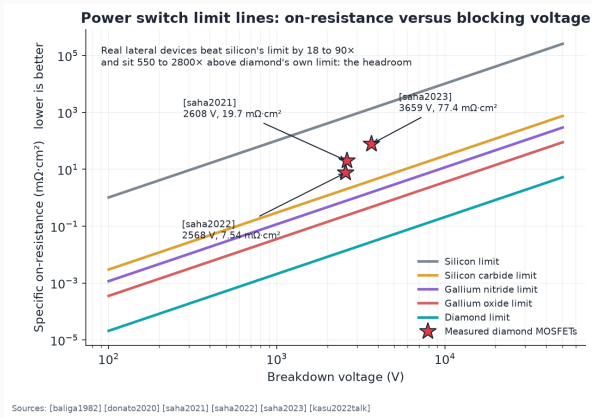
What ADAMAS is

One connected, executable chain, each link checked against published data or an independent tool:

materials → process → circuits → qubits → error correction → machine size → experiments

- 541 references, 500 verified against Crossref
- 52 figures regenerated from code
- Validation matrix: 19 checks, 18 pass
- Process traveler, PDK, placed 4-bit processor
- Circuit-level error correction (Stim, PyMatching)
- Experiments from US\$100, course, 22 web pages

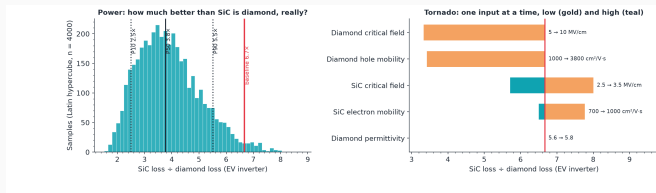
Power switches: real diamond already beats silicon's limit



Best measured: 3659 V; 2022 device about **89×** below silicon's limit.

800 V EV inverter	W/switch
Silicon	209.5
SiC	29.7
GaN	20.6
Diamond (ideal)	3.8

But how sure is that? Uncertainty analysis



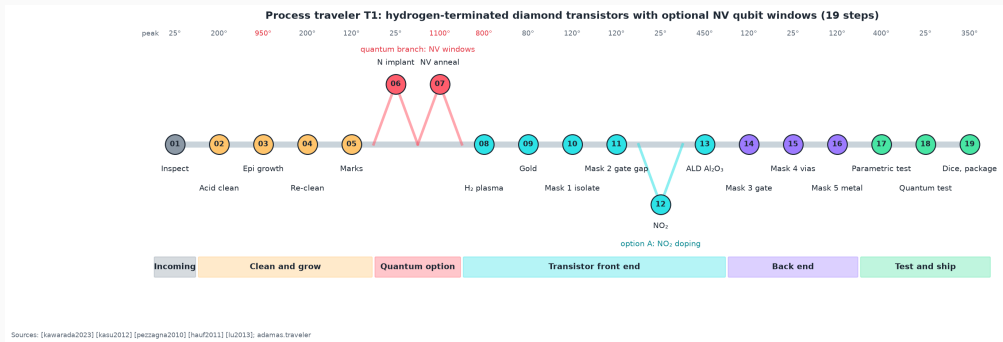
Diamond's critical field (5 to 10 MV/cm) and hole mobility (1,000 to 3,800) are not settled.

SiC ÷ diamond loss: median **3.8×**, P10 to P90 2.5 to 5.5×

Still a big win, but the textbook value is the optimistic end.

Mobility and field each explain ~0.42 of the spread: **measure mobility first** (a Hall setup, about US\$3k).

How you would actually build it



The channel is made by surface chemistry (hydrogen termination), gold protects it, one oxygen plasma isolates every device, and **all heat is spent before the first metal.**

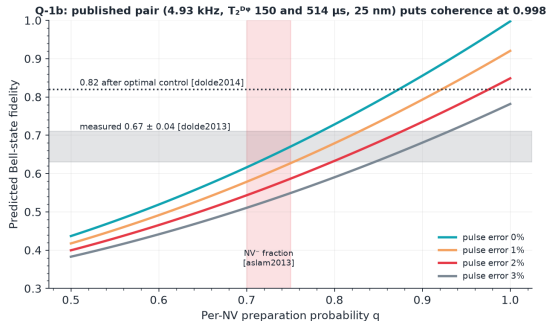
A processor on diamond



Sources: [weste2011] [meadconway1980] [liu2017] [faggin1996] [ajayi2019]; circuits/digital/place.py

- PDK-0: layers, rules, GDS monitor die, standard cells, LEF, timed Liberty
- DIA-4 (4-bit, CALL/RET) synthesizes to 323 cells, about 1 mm² with pads
- Browser emulator matches the Verilog on random programs; ring oscillator within 10% of SPICE

The two-qubit bottleneck is not coherence



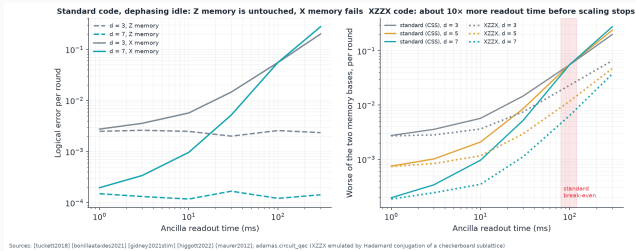
Sources: [dolde2013] [dolde2014] [aslam2013]; adamas.gate_budget model

Published room-temperature pair (25 nm, 25 μs gate): coherence alone allows **0.998**.

Measured 0.67 $\Rightarrow$ preparation $q = 0.75$;
optimal-control 0.82 $\Rightarrow q = 0.87$.

The NV⁻ charge state, not T_2 , limits the gate.

Error correction when readout is slow



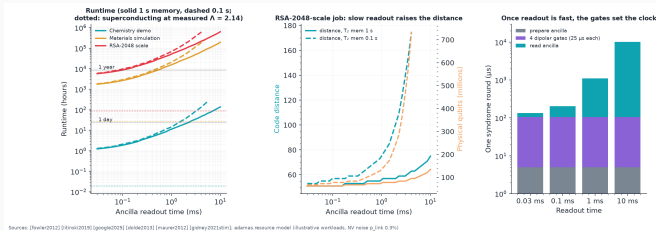
Readout takes about 1 ms while data qubits idle.

$\Lambda = 4.03$ (0.1 ms), 3.56 (1 ms), 2.64 (10 ms).

Scaling stops near 100 ms; the bias-tailored **XZZX** code pushes that to about 1000 ms.

Native circuits with biased gates:
threshold 0.95% $\rightarrow$ 1.59% at
 $\eta = 100$ (standard code: 0.63%).

How big, how slow



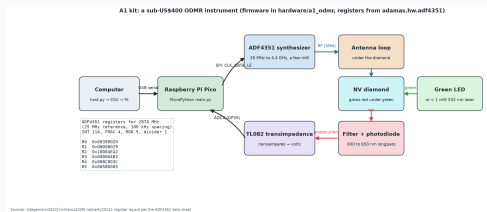
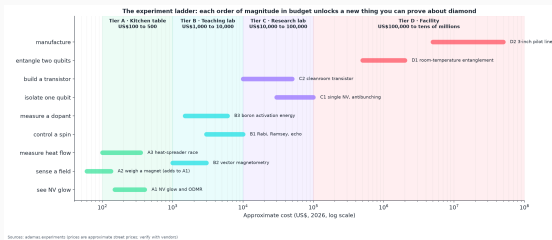
RSA-2048 scale: distance 57,
78 M qubits on a **5.7 mm die**.

Runtime **6.0 years** at 1 ms readout
(1.0 at 0.1 ms), P10 to P90 1.2 to
15.5 years; superconducting: 4
days.

If gate noise is biased, the
standard code balloons to 399 M
qubits; **XZZX keeps it at 67 M**.

Area is free; time is not.

Anyone can start: experiments from US\$100



A1 kit: Pico firmware drives an ADF4351 through the 2.87 GHz resonance; data go straight into the fitter.

Takeaways

1. Diamond power switches are a real advantage: **about $3.8\times$** lower loss than SiC once uncertainty is honest.
2. Room-temperature NV qubits are limited by **charge-state preparation**, not coherence.
3. A room-temperature quantum computer would be **small but slow**; readout speed and bias-tailored codes are the levers.
4. Everything is open, tested, and regenerable: `pytest`, `make figures paper talk`.

github.com/Normansrule/adamas-diamond-framework
normansrule.github.io/adamas-diamond-framework